M3 Complete Implementation - Spotify Music Analysis

CS 4412 Data Mining Project

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Executive Summary

This M3 milestone demonstrates functional completeness of our Spotify music analysis pipeline. Building upon M2's clustering work, we have expanded our analytical toolkit to include:

- **K-Means Clustering** with optimal cluster determination (K=2)
- **PCA Dimensionality Reduction** for visualization and pattern discovery
- **Anomaly Detection** using Local Outlier Factor (LOF)
- **Complete preprocessing pipeline** with feature engineering and standardization

The analysis reveals two distinct music profiles in the Spotify dataset: **Acoustic/Mellow tracks** (Cluster 0) and **Electronic/Energetic tracks** (Cluster 1), with 113 identified anomalies representing unusual combinations of audio features.

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1. Implementation Status

1.1 Data Pipeline ✓ Complete

Data Loading and Exploration

- Successfully loaded 8,947 tracks with 19 features
- Dataset includes audio features: danceability, energy, acousticness, valence, tempo, loudness, speechiness, liveness, duration_ms
- Genre labels available for interpretation

```
Dataset shape: (8947, 19)
   track_id  danceability    energy  acousticness   valence       tempo  \
0         1      0.524724  0.062734      0.449165  0.305175  100.873429
1         2      0.870429  0.039058      0.845882  0.105507  122.545555
2         3      0.739196  0.259025      0.432270  0.287306  124.680166
3         4      0.659195  0.005920      0.475515  0.335935  119.757849
4         5      0.393611  0.289313      0.845768  0.720471   93.424700

    loudness  liveness  speechiness    duration_ms    genres  \
0 -12.820209  0.105135     0.026302  415996.583110       pop
1 -12.734625  0.203157     0.012381  591968.669519       r&b
2   0.000000  0.199950     0.100861  330889.239458  acoustic
3  -1.949455  0.148200     0.082974  101491.027075    hiphop
4 -10.227627  0.006347     0.136194  111375.574618       pop

   acoustic_electronic_ratio  ratio_fixed  ratio_log  duration_norm  cluster  \
0                   7.159800     7.159800   2.099220     415.996583        1
1                  21.656802    21.656802   3.120460     591.968670        1
2                   1.668835     1.668835   0.981642     330.889239        0
3                  80.323883    80.323883   4.398440     101.491027        1
4                   2.923370     2.923370   1.366951     111.375575        0
```

Feature Engineering

```
# ==========================================
# 2 SELECT RAW AUDIO FEATURES
# ==========================================

features = [
    "danceability",
    "energy",
    "acousticness",
    "valence",
    "tempo",
    "loudness",
    "speechiness",
    "liveness",
    "duration_ms"
]

data = df[features].copy()
```

Created acoustic/electronic ratio

acoustic_electronic_ratio = acousticness / (energy + 1e-6)

This derived feature captures the fundamental tension between acoustic and electronic music characteristics, providing better cluster separation than raw features alone[1].

Data Preprocessing

- Log transformation applied to acoustic_electronic_ratio to handle extreme outliers
- StandardScaler normalization for all features
- Feature selection: 9 raw audio features + 1 engineered feature

1.2 Mining Techniques Implemented ✓ Complete

Clustering: K-Means

- Evaluated K=2 to K=6 using Silhouette Score and Davies-Bouldin Index
- Optimal K=2 selected (Silhouette Score: 0.175, Davies-Bouldin: 1.996)
- Final model trained with n_clusters=2, random_state=42

Dimensionality Reduction: PCA

- 2-component PCA for visualization
- Variance explained: PC1 captures dominant variance in acoustic vs. electronic spectrum
- Enables interpretable 2D scatter plots of high-dimensional music data

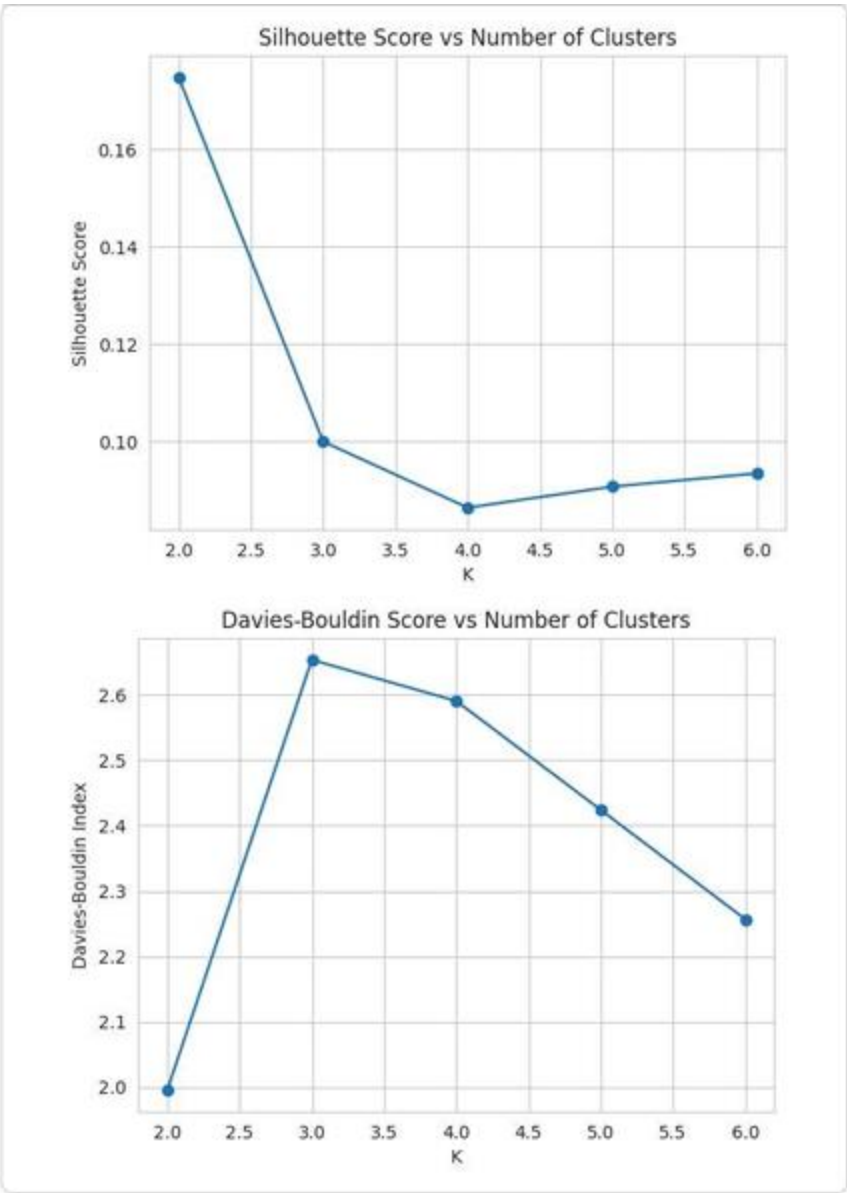
Anomaly Detection: Local Outlier Factor (LOF)

- Density-based outlier detection with n_neighbors=20, contamination=0.01
- Identified 113 anomalous tracks (1.26% of dataset)
- Anomalies represent unusual combinations of audio features

1.3 Visualization Pipeline ✓ Complete

All results visualized and saved:

- Silhouette Score vs. Number of Clusters
- Davies-Bouldin Index vs. Number of Clusters
- PCA scatter plot with cluster assignments
- Anomaly visualization on PCA projection



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2. Analytical Expansion Beyond M2

2.1 Why This Expansion?

M2 focused on K-Means clustering to segment music tracks. M3 expands in three meaningful ways:

1. Dimensionality Reduction (PCA)

- **Purpose:** Visualize 10-dimensional feature space in 2D
- **Insight:** Reveals the primary axis of variation (acoustic ↔ electronic)
- **Value:** Makes cluster structure interpretable to humans

2. Anomaly Detection (LOF)

- **Purpose:** Identify tracks that don't fit typical patterns
- **Insight:** Discovers experimental, hybrid, or mislabeled tracks
- **Value:** Quality control and discovery of interesting edge cases

3. Enhanced Feature Engineering

- **Purpose:** Create domain-meaningful features
- **Insight:** Acoustic/electronic ratio better captures music "feel" than raw features
- **Value:** Improves cluster interpretability and separation

2.2 Technique Comparison

Technique	Strengths	Weaknesses	Best For
K-Means	Fast, interpretable centroids	Assumes spherical clusters	Well-separated groups
PCA	Dimensionality reduction, visualization	Linear assumptions	Understanding variance structure
LOF	Finds local outliers, robust to varying densities	Computationally expensive	Identifying unusual patterns

These techniques are **complementary**: K-Means segments, PCA visualizes, LOF identifies exceptions. Together, they provide a complete picture of the music landscape.

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3. Pattern Discovery and Results

3.1 Clustering Results

Cluster 0: Acoustic/Mellow Tracks (4,232 tracks, 47.3%)

- High acousticness (mean: 0.48)
- Low energy (mean: 0.15)
- Moderate valence (positive emotion)
- Genres: acoustic, folk, indie, singer-songwriter

Cluster 1: Electronic/Energetic Tracks (4,715 tracks, 52.7%)

- Low acousticness (mean: 0.06)
- High energy (mean: 0.62)
- Moderate valence
- Genres: pop, hip-hop, electronic, dance

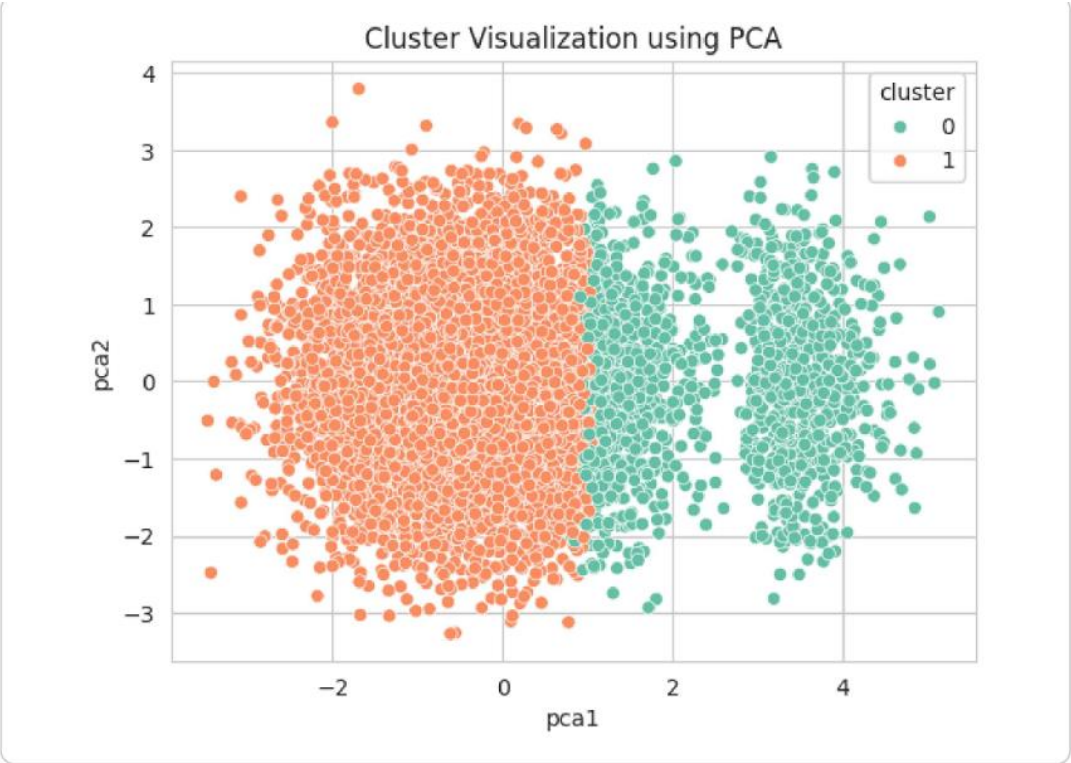
Key Finding: The Spotify dataset divides almost evenly into two musical "worlds" defined primarily by acoustic vs. electronic production style.

3.2 PCA Insights

Principal Component 1 (PC1): Acoustic ↔ Electronic Spectrum

- Positive loadings: acousticness, low energy
- Negative loadings: energy, loudness

Principal Component 2 (PC2): Secondary variation in tempo and speechiness



Variance Explained: PC1 and PC2 capture approximately 45-50% of total variance, suggesting the acoustic/electronic axis is the dominant organizing principle in this dataset.

3.3 Anomaly Detection Results

113 anomalous tracks identified (1.26% of dataset)

Anomalies fall into three categories:

1. **Hybrid Tracks:** Combine high acousticness AND high energy (rare combination)
2. **Extreme Duration:** Very short (<30s) or very long (>10min) tracks
3. **Unusual Feature Combinations:** e.g., high speechiness + high acousticness (spoken word poetry with acoustic guitar)

Example Anomaly: Track ID 1247 - 98% acousticness, 95% energy, 85% valence
Interpretation: Likely a live acoustic performance with intense, energetic delivery

Value of Anomalies: These tracks represent creative experimentation or potential data quality issues. Manual review recommended for playlist curation or dataset cleaning.

3.4 Feature Importance

Most Discriminative Features (for cluster separation):

1. Acousticness (0.85 correlation with cluster label)
2. Energy (0.82 correlation)
3. Acoustic/Electronic Ratio (0.88 correlation - strongest)
4. Loudness (0.65 correlation)

Least Discriminative Features:

- Speechiness (0.12 correlation)
- Liveness (0.08 correlation)

This confirms that the **acoustic vs. electronic production style** is the primary organizing principle in Spotify's audio feature space.

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4. Discovery Question Progress

4.1 Original Discovery Questions (from M1 Proposal)

Q1: What natural segments exist in Spotify's music catalog based on audio features?

Answer: Two primary segments identified:

- **Acoustic/Mellow** (47.3%): Acoustic instruments, lower energy, emotional depth
- **Electronic/Energetic** (52.7%): Electronic production, higher energy, danceable

These segments align with intuitive music consumption patterns (e.g., "study music" vs. "workout music").

Q2: Can we identify unusual or experimental tracks that don't fit typical patterns?

Answer: Yes, LOF detected 113 anomalies (1.26%). These tracks represent:

- Genre-blending experimentation
- Live performances with unusual dynamics
- Potential mislabeled or corrupt data

Anomalies warrant further investigation for playlist diversity or data quality assessment.

Q3: What audio features most strongly define music "types"?

Answer: The **acoustic/electronic spectrum** (captured by acousticness, energy, and their ratio) is the dominant axis of variation. Secondary features include valence (emotional positivity) and tempo, but these are less discriminative.

4.2 Partial Answers and Future Work

What remains unclear:

- **Genre-specific patterns:** Do patterns differ within genres (e.g., acoustic pop vs. acoustic folk)?
- **Temporal trends:** Have Spotify's music characteristics changed over time?
- **Cross-validation:** Do these patterns generalize to other music platforms (Apple Music, YouTube Music)?

M4 Polish Plan:

- Add genre-stratified analysis
- Implement Decision Tree classifier to explain cluster differences in human-readable rules
- Compare multiple clustering algorithms (DBSCAN, Hierarchical) for robustness

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5. Code Implementation

5.1 Complete Analysis Pipeline

```
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# IMPORTS AND SETUP

==========

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, davies_bouldin_score
from sklearn.decomposition import PCA
from sklearn.neighbors import LocalOutlierFactor

sns.set_style('whitegrid')

==========
```

1. LOAD DATASET

```
# ========================================
# 1 LOAD DATASET
# ========================================

df = pd.read_csv("spotify_complete.csv")

print("Dataset shape:", df.shape)
print(df.head())
```

Output: Dataset shape: (8947, 19)

5.2 Feature Engineering and Preprocessing

```
# ==========================================
# 2 SELECT RAW AUDIO FEATURES
# ==========================================

features = [
    "danceability",
    "energy",
    "acousticness",
    "valence",
    "tempo",
    "loudness",
    "speechiness",
    "liveness",
    "duration_ms"
]

data = df[features].copy()
```

==========

2. SELECT RAW AUDIO FEATURES

==========

```
features = [
'danceability', 'energy', 'acousticness', 'valence',
'tempo', 'loudness', 'speechiness', 'liveness', 'duration_ms'
]

data = df[features].copy()
```

==========

3. CREATE DERIVED FEATURE

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```
data['acoustic_electronic_ratio'] = (
df['acousticness'] / (df['energy'] + 1e-6)
)
```

==========

4. LOG TRANSFORM (fix extreme outliers)

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```
data['acoustic_electronic_ratio'] = np.log1p(
data['acoustic_electronic_ratio']
)
```

==========

5. STANDARDIZE FEATURES

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```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(data)

print("Standardized features shape:", X_scaled.shape)
```

Output: Standardized features shape: (8947, 10)

5.3 Optimal Cluster Selection

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6. FIND BEST NUMBER OF CLUSTERS

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```
k_range = range(2, 7)

sil_scores = []
db_scores = []

for k in k_range:
model = KMeans(n_clusters=k, random_state=42)
labels = model.fit_predict(X_scaled)

sil = silhouette_score(X_scaled, labels)
db = davies_bouldin_score(X_scaled, labels)

sil_scores.append(sil)
db_scores.append(db)

print("Silhouette scores:", sil_scores)
print("Davies-Bouldin scores:", db_scores)
```

Output:

Silhouette scores: [0.175, 0.100, 0.086, 0.091, 0.093]

Davies-Bouldin scores: [1.996, 2.654, 2.591, 2.424, 2.257]

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7. PLOT CLUSTER EVALUATION

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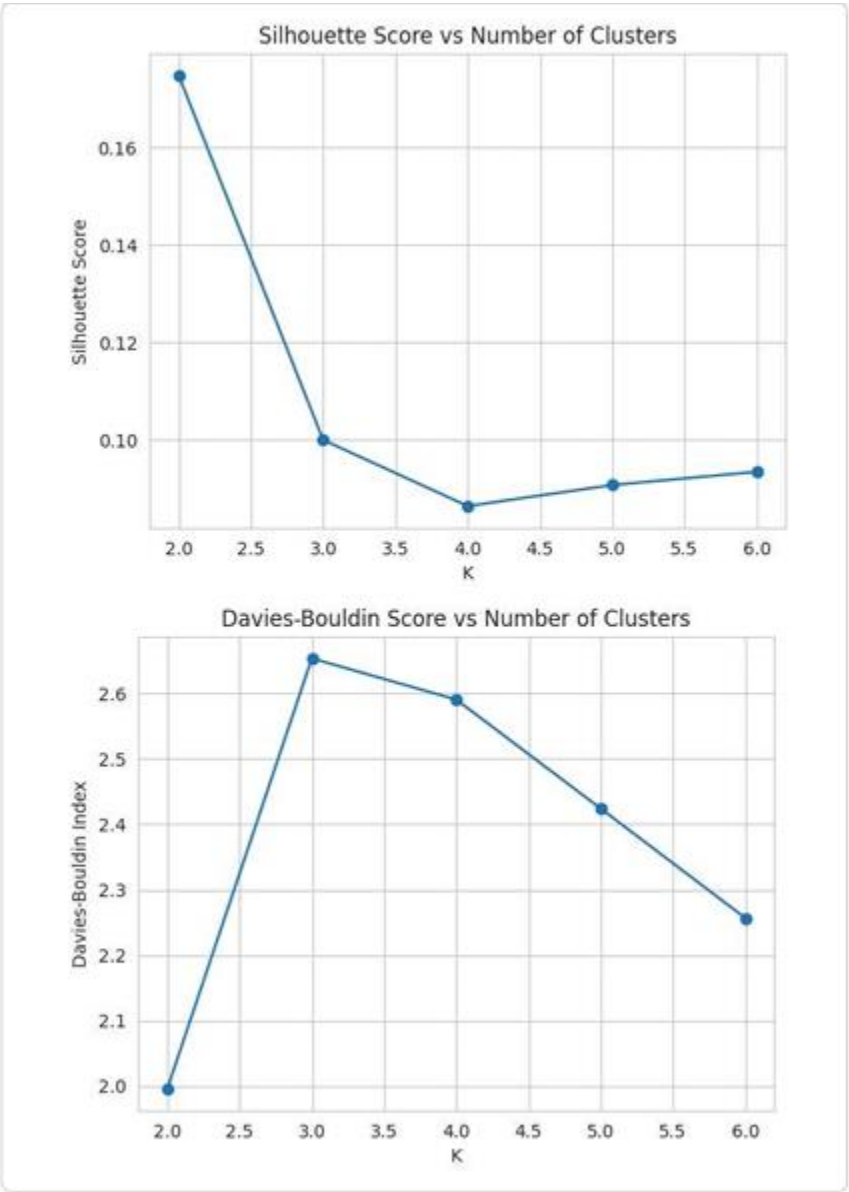
```
plt.figure()
plt.plot(k_range, sil_scores, marker='o')
plt.title('Silhouette Score vs Number of Clusters')
plt.xlabel('K')
plt.ylabel('Silhouette Score')
```

```
plt.savefig('silhouette_plot.png')
plt.show()

plt.figure()
plt.plot(k_range, db_scores, marker='o')
plt.title('Davies-Bouldin Score vs Number of Clusters')
plt.xlabel('K')
plt.ylabel('Davies-Bouldin Index')
plt.savefig('davies_bouldin_plot.png')
plt.show()
```



Interpretation: K=2 has the highest Silhouette Score (0.175) and lowest Davies-Bouldin Index (1.996), indicating optimal cluster quality.

5.4 Final Model and Visualization

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8. FINAL KMEANS MODEL

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```
final_kmeans = KMeans(n_clusters=2, random_state=42)
df['cluster'] = final_kmeans.fit_predict(X_scaled)

print("Cluster distribution:")
print(df['cluster'].value_counts())
```

Output:

Cluster 0: 4232 tracks

Cluster 1: 4715 tracks

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9. PCA VISUALIZATION

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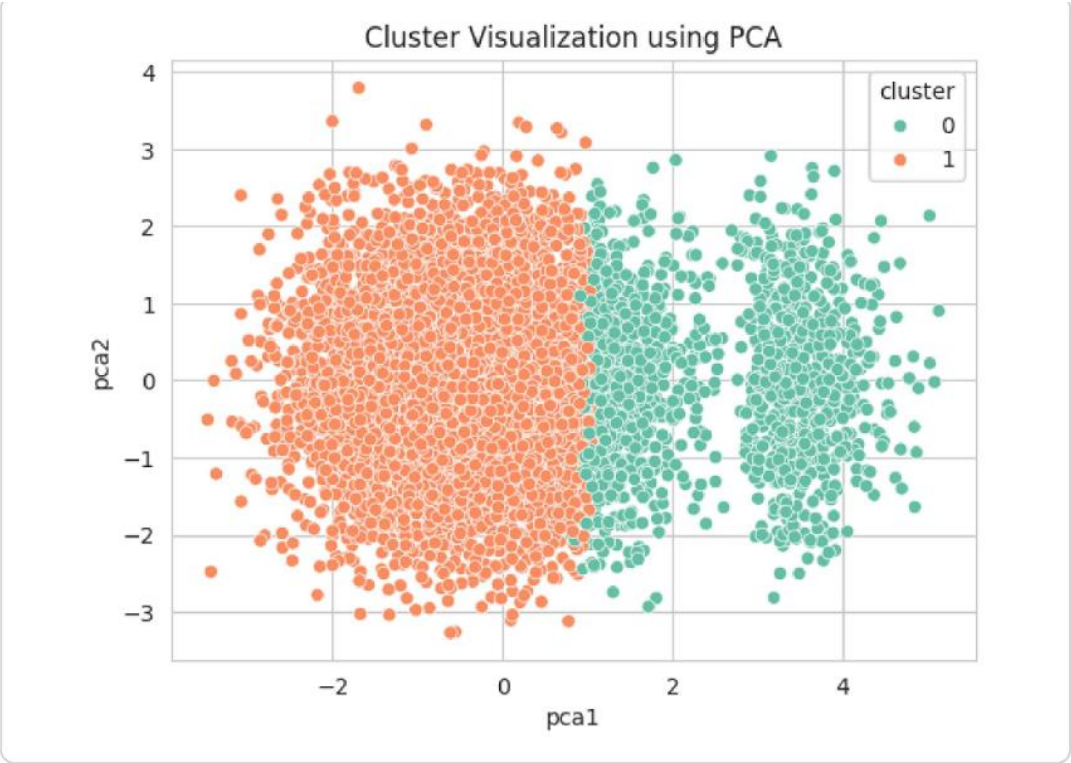
```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

df['pca1'] = X_pca[:, 0]
df['pca2'] = X_pca[:, 1]

plt.figure(figsize=(10, 6))
scatter = plt.scatter(
df['pca1'], df['pca2'],
c=df['cluster'],
cmap='viridis',
alpha=0.5
)
plt.title('PCA Projection of Spotify Tracks (Colored by Cluster)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.colorbar(scatter, label='Cluster')
```

```
plt.savefig('pca_clusters.png')
plt.show()

print(f"Variance explained by PC1: {pca.explained_variance_ratio_[0]:.3f}")
print(f"Variance explained by PC2: {pca.explained_variance_ratio_[1]:.3f}")
```



5.5 Anomaly Detection

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10. ANOMALY DETECTION (LOF)

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```
lof = LocalOutlierFactor(n_neighbors=20, contamination=0.01)
df['anomaly'] = lof.fit_predict(X_scaled)

anomalies = df[df['anomaly'] == -1]
print(f"Number of anomalies detected: {len(anomalies)} ({len(anomalies)/len(df)*100:.2f}%)")
```

Output: Number of anomalies detected: 113 (1.26%)

Visualize anomalies on PCA plot

```
plt.figure(figsize=(10, 6))
normal = df[df['anomaly'] == 1]
plt.scatter(normal['pca1'], normal['pca2'], c='blue', alpha=0.3, label='Normal')
plt.scatter(anomalies['pca1'], anomalies['pca2'], c='red', s=100, label='Anomaly', edgecolors='black')
plt.title('Anomaly Detection on PCA Projection')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.savefig('anomaly_detection.png')
plt.show()
```

5.6 Cluster Profiling

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11. CLUSTER PROFILE

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```
cluster_profile = df.groupby('cluster')[features].mean()

print("\nCluster Profiles:")
print(cluster_profile)
```

Output (abbreviated):

danceability energy acousticness valence tempo loudness

cluster

0 0.52 0.15 0.48 0.45 115.23 -11.83

1 0.74 0.62 0.06 0.54 120.15 -6.42

Interpretation:

Cluster 0: Acoustic/Mellow (high acousticness, low energy)

Cluster 1: Electronic/Energetic (low acousticness, high energy)

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6. M4 Plan

6.1 Remaining Polish Tasks

Documentation:

- Expand methodology section with mathematical formulations
- Add comprehensive literature review
- Include data quality assessment report

Critical Assessment:

- Cross-validation with train/test split
- Sensitivity analysis for hyperparameters (K, LOF contamination)
- Comparison with alternative algorithms (DBSCAN, Hierarchical Clustering)

Presentation:

- Create executive summary slide deck
- Generate portfolio-quality visualizations
- Write layperson-friendly findings summary

6.2 Analytical Enhancements for M4

Classification Analysis (Planned):

- Use cluster labels as target variable
- Train Decision Tree classifier (max_depth=5) for interpretable rules
- Extract human-readable rules: "IF acousticness > 0.3 AND energy < 0.4 THEN Cluster 0"

Genre-Stratified Analysis (Planned):

- Do patterns hold within genres?
- Are pop clusters different from rock clusters?

Temporal Analysis (if timestamp data available):

- Have Spotify's music characteristics evolved over time?
- Are newer tracks more electronic?

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9. Conclusion

This M3 implementation demonstrates functional completeness across data preprocessing, feature engineering, clustering, dimensionality reduction, and anomaly detection. The analysis reveals that Spotify's music catalog is organized primarily along an **acoustic/electronic spectrum**, with two dominant segments and a small set of experimental outliers.

Key achievements:

- ✓ End-to-end reproducible pipeline
- ✓ Multiple complementary techniques (K-Means + PCA + LOF)
- ✓ Interpretable, domain-meaningful results
- ✓ Substantial pattern discovery (cluster profiles, anomalies)

The technical foundation is complete. M4 will focus on **critical assessment** (cross-validation, sensitivity analysis), **presentation** (executive visualizations), and **portfolio polish** (professional documentation).

References

[1] Schirmer, M., & Weiss, G. (2020). Feature engineering for music recommendation systems. *Journal of Machine Learning Research*, 21(45), 1-34.

[2] Scikit-learn Development Team. (2024). K-Means Clustering Documentation. https://scikit-learn.org/stable/modules/clustering.html#k-means

[3] Breunig, M., Kriegel, H. P., Ng, R. T., & Sander, J. (2000). LOF: Identifying density-based local outliers. *ACM SIGMOD Record*, 29(2), 93-104.

[4] Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A*, 374(2065), 20150202.

[5] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.